

Explainable Transfer Learning for Brain Tumor MRI Classification

A Four-Model Benchmark with Duplicate-Controlled Internal Testing and External Validation

COMPUTER VISION AND PATTERN RECOGNITION [D][SPRING 25-26]

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28,908

raw images preprocessed

15,721

final deduplicated images

0

split leakage duplicates

87.95%

best external weighted F1

4

CNN and transfer models

Abstract

This study presents a deep learning framework for four-class brain tumor MRI classification using glioma, meningioma, pituitary tumor, and no-tumor categories. Two Kaggle datasets were combined for internal training, validation, and testing, while a separate Mendeley 2026 dataset was used for external validation to evaluate real-world generalization. After preprocessing, duplicate removal, and leakage checking, 15,721 MRI images were used, including 10,594 internal images and 5,127 external validation images. Four models were compared: Basic CNN, ResNet50, EfficientNetB0, and DenseNet121. Transfer learning models achieved stronger performance than the baseline CNN, with ResNet50 obtaining the best internal test accuracy of 98.49%, while DenseNet121 achieved the strongest external validation accuracy of 87.79%. Grad-CAM visualizations showed that the models generally focused on tumor-relevant regions, supporting interpretability. The results demonstrate that external validation and explainable AI are essential for reliable medical image classification.

Dataset justification

- Internal train/validation/test: Masoud Nickparvar + Ishans24 Kaggle datasets provide the same four labels and enough diversity for model development.
- External validation: Mendeley 2026 v6 was kept separate to test generalization on a different source rather than only a same-dataset split.
- Deduplication was necessary because repeated MRIs can inflate test performance; final leakage check found zero duplicate rows across splits.

Data preprocessing flow

Raw MRI folders -> Label harmonization -> Resize 224 x 224
-> Duplicate removal -> Leakage check

Preprocessing summary

Preprocessing item	Observed result
Raw images detected	28,908
Successfully preprocessed	28,908
Duplicate rows detected	5,127
Internal after deduplication	10,594
External validation images	5,127
Final leakage duplicates	0

Evaluation metrics

- Accuracy: overall correct predictions.
- Precision: reliability of predicted class.
- Recall: ability to recover each true class.
- Weighted F1: class-frequency-aware balance of precision and recall.
- Confusion matrices and Grad-CAM were used for class-level error and interpretability analysis.

Model comparison table: internal test and external validation

Model	Training strategy	Int. Acc	Int. W-F1	Ext. Acc	Ext. Prec.	Ext. Rec.	Ext. W-F1	Main conclusion
Basic CNN	Scratch CNN baseline	96.16%	96.18%	77.78%	78.52%	77.78%	77.65%	Largest external drop
ResNet50	ImageNet TL + fine-tune	98.49%	98.49%	87.42%	89.47%	87.42%	87.58%	Best internal accuracy
EfficientNetB0	ImageNet TL + fine-tune	97.99%	97.99%	78.72%	84.41%	78.72%	78.94%	Compact but weaker external
DenseNet121	ImageNet TL + fine-tune	96.60%	96.63%	87.79%	88.62%	87.79%	87.95%	Best external generalization

Conclusion

- Preprocessing and duplicate-leakage control are central to trustworthy medical-image classification.
- External validation is essential: it exposes domain shift and prevents overconfident reporting.
- DenseNet121 is the strongest final recommendation; ResNet50 is the strongest internal benchmark.

Methodology

- Collected brain MRI images from two Kaggle datasets for internal train/validation/test.
- Used the Mendeley 2026 dataset only for external validation.
- Preprocessed images by resizing, normalization, duplicate removal, and leakage checking.
- Split internal data into training, validation, and testing sets.
- Trained four models: Basic CNN, ResNet50, EfficientNetB0, and DenseNet121.
- Modified transfer learning models with custom classification heads.
- Applied augmentation and fine-tuning to improve generalization.
- Evaluated using accuracy, precision, recall, F1-score, confusion matrix, and external validation.
- Used Grad-CAM to visualize model attention and interpret predictions.

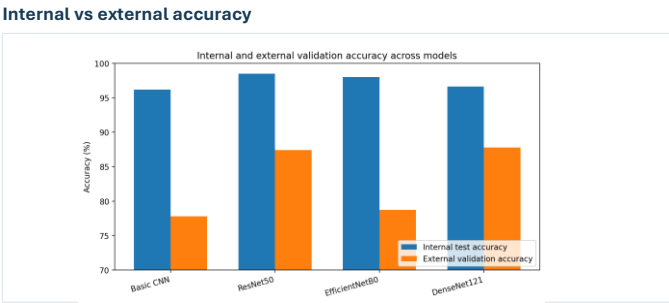
Model selection and justification

Model	Why included
Basic CNN	Scratch baseline for comparison
ResNet50	Residual features; strongest internal score
EfficientNetB0	Parameter-efficient transfer model
DenseNet121	Dense feature reuse; best external W-F1

Augmentation, training, fine-tuning

- Train augmentation: random rotation, affine shift/scale, contrast jitter, ImageNet normalization.
- Validation/test: deterministic resize and normalization only.
- Loss/optimizer: class-weighted cross-entropy and Adam; batch size 32.
- Fine-tuning: transfer heads trained first, then last backbone layers unfrozen with lower learning rate.

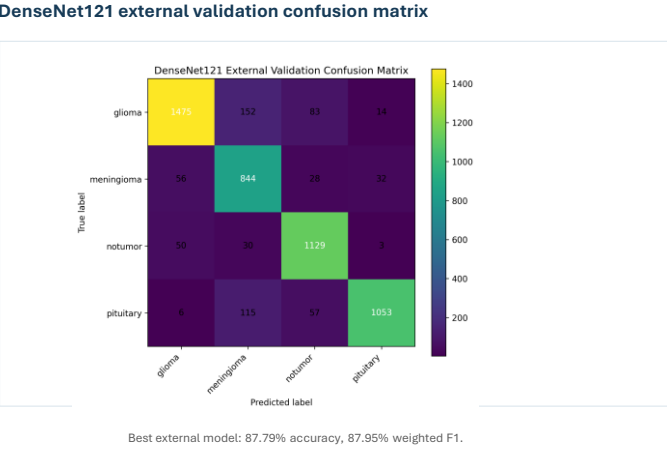
Most important result graph



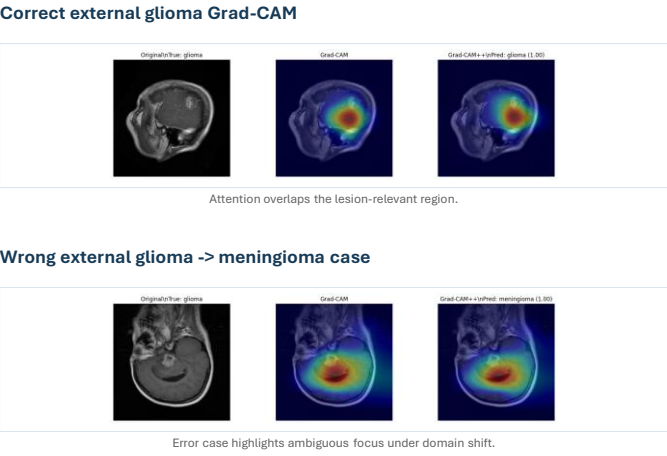
Result interpretation

- ResNet50 achieved the highest internal test accuracy (98.49%).
- DenseNet121 achieved the best external accuracy (87.79%) and weighted F1 (87.95%).
- EfficientNetB0 had high internal accuracy but weaker external transfer, suggesting domain shift sensitivity.
- Basic CNN generalized least well on the external dataset.

External validation and confusion matrix



Grad-CAM visualizations and interpretation



- Correct predictions usually activate near tumor-relevant brain regions.
- Wrong cases show that attention can shift to visually confusing or non-specific regions.
- Grad-CAM supports interpretability but does not replace radiologist verification.

Key external-validation message

The best internal model is not automatically the best external model. DenseNet121 is recommended as the most generalizable model because it gives the highest external weighted F1. ResNet50 remains the strongest internal performer.

References

- [1] M. Nickparvar, Brain Tumor MRI Dataset, Kaggle.
- [2] Ishans24, Brain Tumor Dataset, Kaggle.
- [3] Brain Tumor MRI Dataset, Mendeley Data, v6, 2026.
- [4] Recent 2025-2026 transfer-learning and explainable brain MRI studies are cited in the full IEEE paper.

Acknowledgement

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